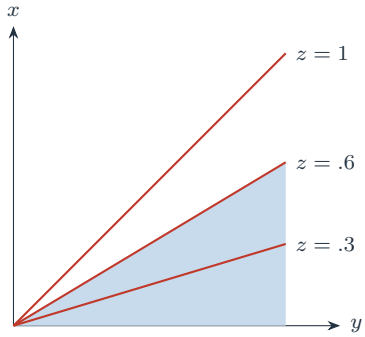
Условие

Нека $X \sim \mathrm{Exp}(2)$ и $Y \sim \mathrm{Unif}(0, 3)$ са независими. Намерете разпределението на $Z = X/Y$.

Идея: области за $X/Y \leq z$

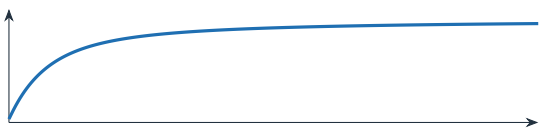


под правата е областта $x \leq zy$

Обусловяме по X

$\mathbb{P}(X \le zY) = \int \mathbb{P}(Y \ge \frac{x}{z}) f_X(x)dx$, като $\mathbb{P}(Y \ge \frac{x}{z}) = \frac{3-x/z}{3}$ при $x < 3z$.

$$F_Z(z) = \int_0^{3z} 2\frac{3-\frac{x}{z}}{3} e^{-2x} dx = 1 - \frac{1}{6z} + \frac{e^{-6z}}{6z}.$$



Отговор. $f_Z(z) = \dfrac{1-(1+6z)e^{-6z}}{6z^2} \mathbb{1}_{(0,\infty)}(z).$